

$$\begin{aligned}
& (\sigma_0 E[(\text{async}/\text{lambda } (x \dots) e) v \dots])) \longrightarrow (\sigma_1 E[(\text{coro } (\text{lambda } (x_{dummy}) \text{ lib:begin}[\![x_{dummy}, e]\!])])]) \quad [\text{async-app}] \\
& \text{where } x_{dummy} = \text{gensym}[\![\sigma_0, \text{dummy}]\!], \sigma_1 = \text{ext}[\![\sigma_0, (x \ v), \dots]\!] \\
& (\sigma_0 E[(\text{tagged } x_{running} E_{inner}[(\text{await } (\text{tag } x_{coro}))])]) \longrightarrow (\sigma_1 E[(\text{tag } x_{running})]) \quad [\text{await-coro}] \\
& \text{where } (\text{not in-tag?}[\![E_{inner}\!])], x_{dummy} = \text{gensym}[\![\sigma_0, \text{dummy}]\!], \\
& \quad v_{coro} = (\text{coroutine} \\
& \quad \quad (\text{lambda } (x_{dummy}) \\
& \quad \quad \quad E_{inner} \\
& \quad \quad \quad [\text{lib:begin}[\![x_{dummy}, \\
& \quad \quad \quad \quad (\text{await } (\text{resume! } (\text{tag } x_{coro}) (\text{void})))\!]])]) \\
& \quad \sigma_1 = \text{ext1}[\![\sigma_0, (x_{running} \ v_{coro})]\!] \\
& (\sigma E[(\text{await } v)]) \longrightarrow (\sigma E[v]) \quad [\text{await-continue}] \\
& \quad \text{where } (\text{not awaitable?}[\![v]\!])
\end{aligned}$$